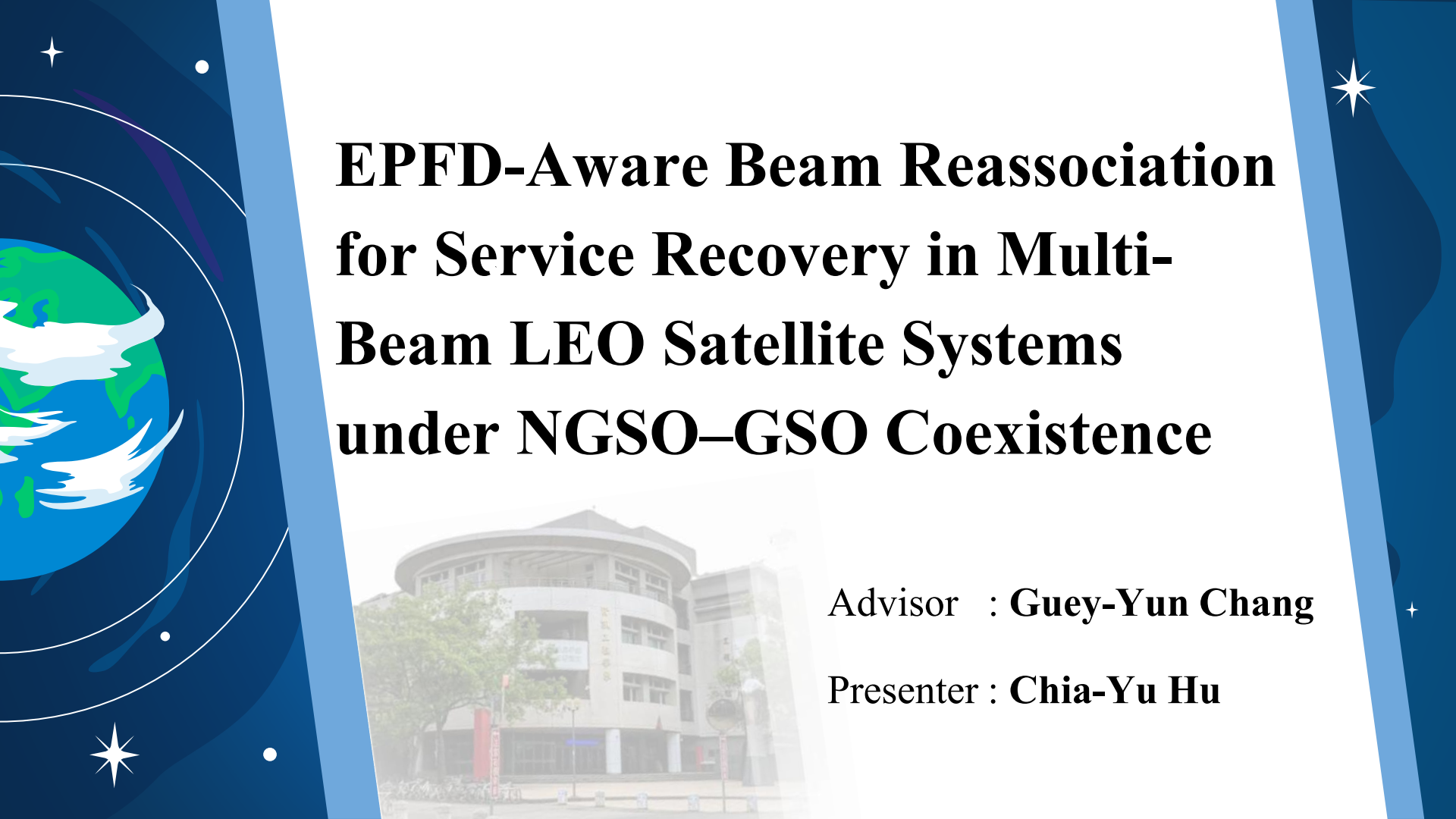




EPFD-Aware Beam Reassociation for Service Recovery in Multi- Beam LEO Satellite Systems under NGSO–GSO Coexistence



Advisor : **Guey-Yun Chang**

Presenter : **Chia-Yu Hu**

Outline

- 1. Introduction**
- 2. Related Work**
- 3. System Model and Problem Formulation**
- 4. Method**
- 5. Evaluation**
- 6. Conclusion**

Introduction — Motivation

- LEO (NGSO) satellite constellations are being widely deployed to provide stable and fast global internet services.
- LEO satellites provide lower latency and more flexible coverage than traditional GSO satellites.
- The increasing number of LEO satellites leads to denser use of orbital and spectrum resources.
- Spectrum sharing becomes an important issue in NGSO–GSO coexistence.

Introduction — NGSO–GSO Coexistence

- **NGSO–GSO Co-Channel Interference:**
 - NGSO and GSO systems may share the same frequency bands.
 - NGSO downlink transmissions can create co-channel interference toward GSO earth stations.
 - Interference becomes critical when an NGSO satellite is close to the GSO direction observed by the GS.
 - ITU Radio Regulations Article 22 defines EPFD limits to protect GSO systems from

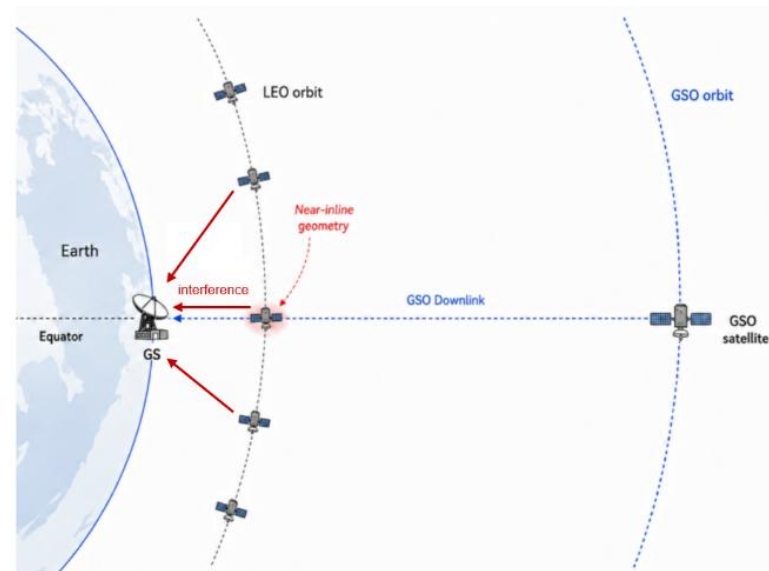


Fig. 1. NGSO–GSO coexistence scenario and near-inline interference geometry between a LEO satellite, a GSO ground station, and the target GSO satellite.

Related Work

	Beam shot down	Power control	Tilt angle	Beamforming
Progressive Pitch Strategy <i>Li et al.</i> _[1]				
Power and Tilt Control <i>Jalali et al.</i> _[2]				
Beamforming <i>Jalali et al.</i> _[3]				

[1] *Research on interference avoidance effect of OneWeb satellite constellation's progressive pitch strategy*, (2021, *International Journal of Satellite Communications and Networking*)

[2] *Joint Power and Tilt Control in Satellite Constellation for NGSO-GSO Interference Mitigation*, (2023, *IEEE Communications*)

[3] *NGSO Satellites Beam Hopping Strategy Based on Load Balancing and Interference Avoidance for Coexistence With GSO Systems*, (2023, *International Journal of Satellite Communications and Networking*) **Interference Mitigation**

Related Work - Research Gap

- **Research Gap**

- These mitigation actions may reduce the available resources for LEO users served by the affected beams.
- Continuous power control, attitude adjustment, or beamforming may require frequent updates under fast LEO satellite motion.
- EPFD-Aware Beam Reassociation(EABR) for Service uses overlapping beam footprints among neighboring LEO satellites to recover affected users while satisfying the EPFD constraint.

System Model — Network Scenario

- A multi-beam LEO satellite system is considered under NGSO–GSO coexistence.
- The GS communicates with its target GSO satellite, and the GSO–GS direction is used as the reference direction for EPFD evaluation.
- The aggregate EPFD is the total interference contribution received at the GS from all beams of visible LEO satellites.
- When a visible LEO satellite covers the GS region and passes close to the GSO–GS direction, the aggregate EPFD may exceed the required limit.

System Model — Beam Power Model

- Under normal operation, each beam on a LEO satellite uses a preconfigured default transmit power.
- Each beam has a maximum transmit power limit.
- Each satellite has a total transmit power budget.

System Model — User Reassociation Setting

- Each user has only one serving beam in each time slot.
- A beam can serve a user only if its footprint covers that user.
- When multiple beams from different LEO satellites cover the same user, one of them can be selected as the serving beam.
- The feasible beams are assumed to use compatible downlink frequency and polarization settings.
- Therefore, user reassociation means changing the serving beam to another feasible beam.

System Model — Satisfaction Model

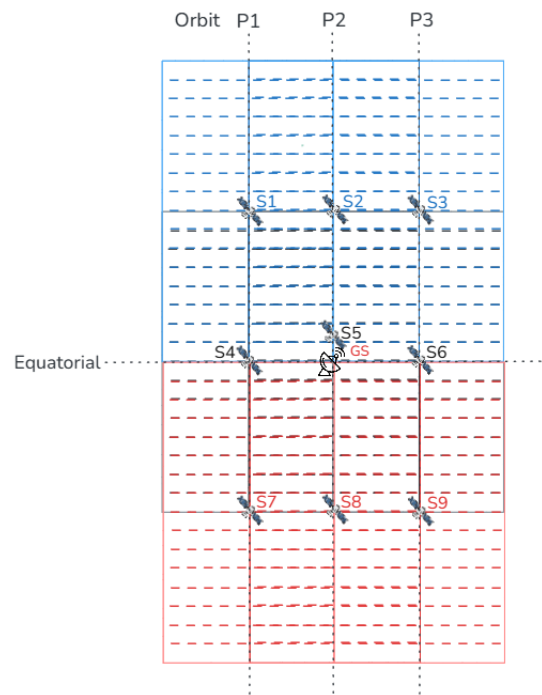
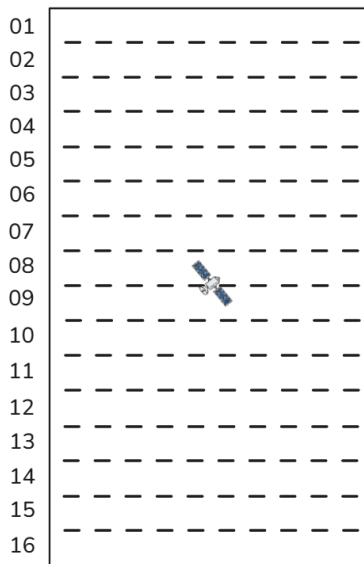
- Users served by the same beam share the available downlink resource.
- The service rate of each user is determined by the achievable capacity and beam loading.
- User satisfaction measures how much of the user demand is satisfied.
- Each user is assigned a service priority weight based on the 3GPP QoS priority concept.

Method — Design Motivation

- Goal: improve overall user satisfaction under the EPFD limit.
- Key idea: use overlapping beam footprints among neighboring LEO satellites to reassociate affected users to feasible serving beams.

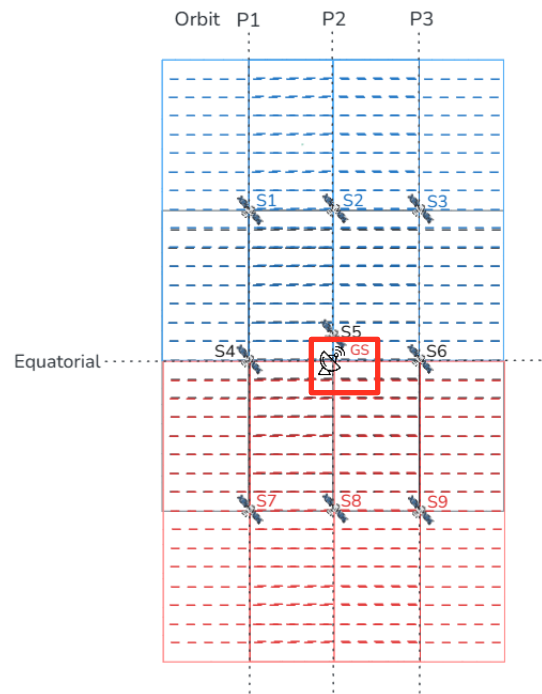
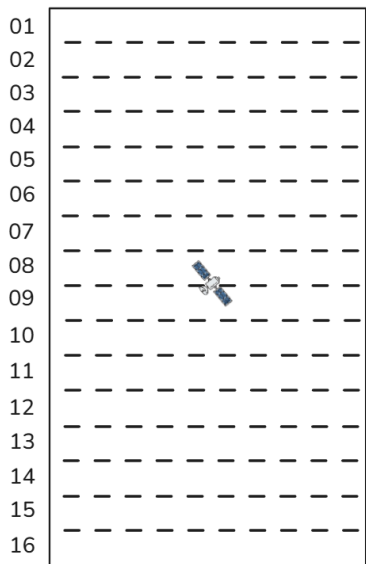
Method — Design Motivation

- OneWeb-like multi-beam is used as an example to illustrate the beam overlap and reassociation relations.



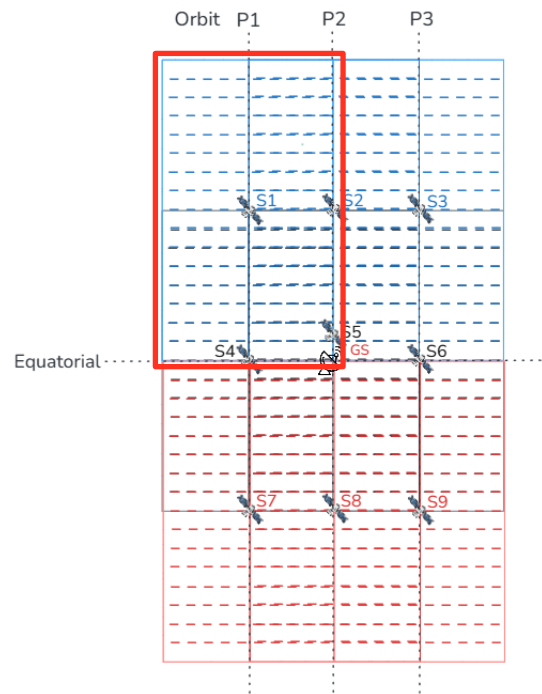
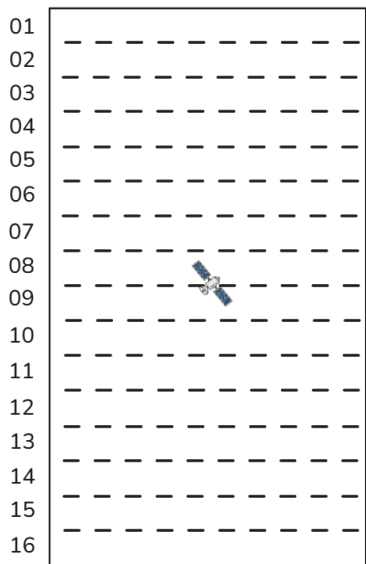
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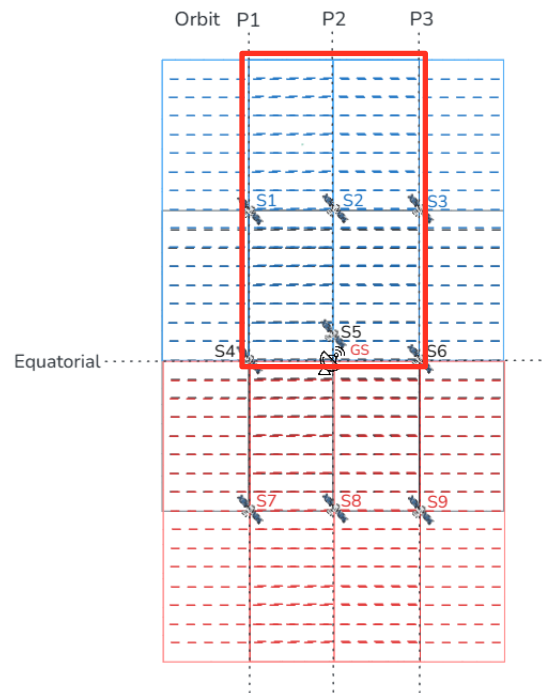
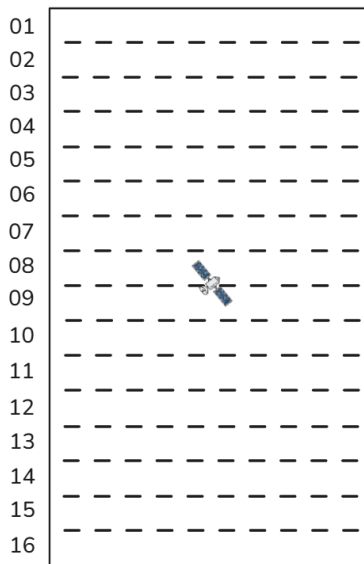
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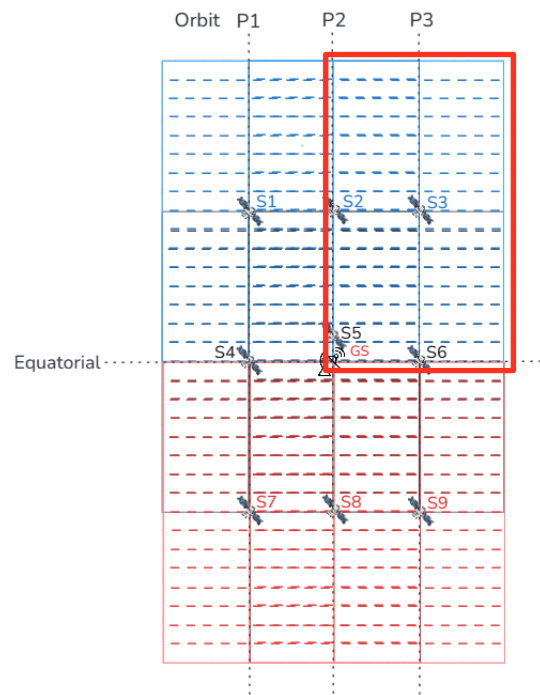
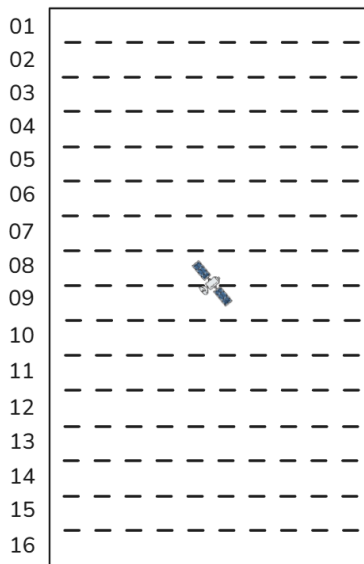
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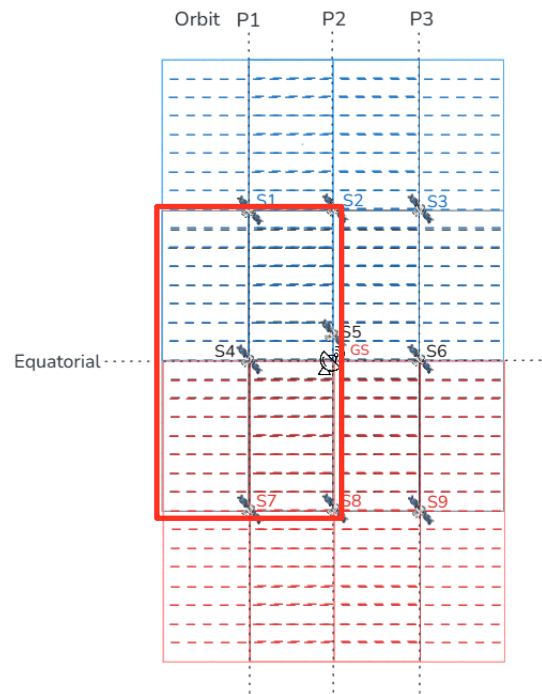
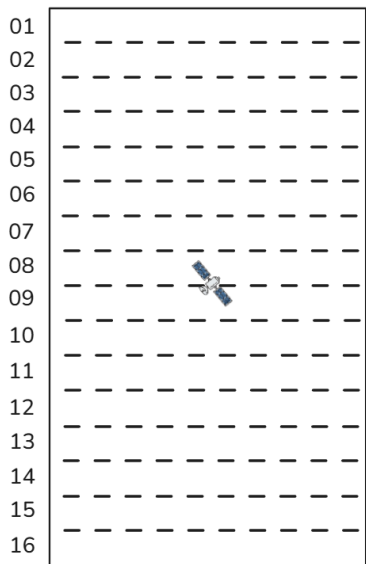
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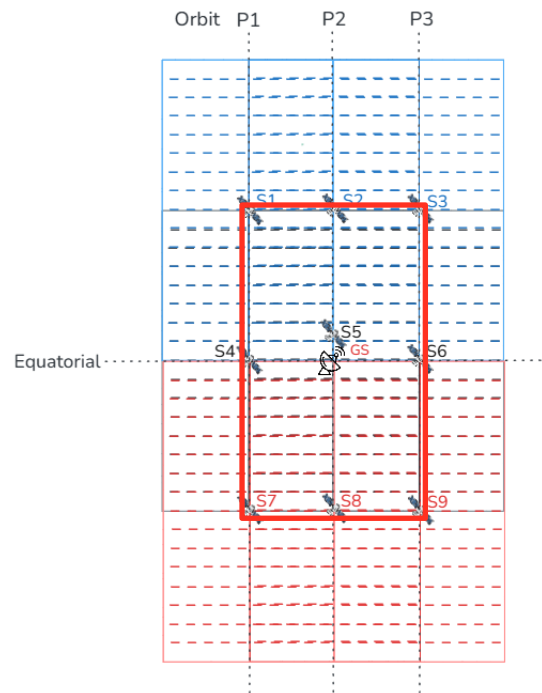
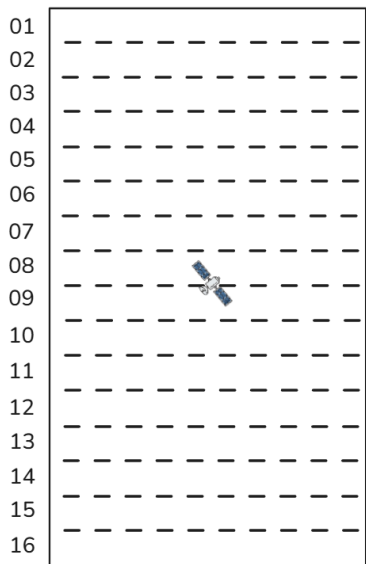
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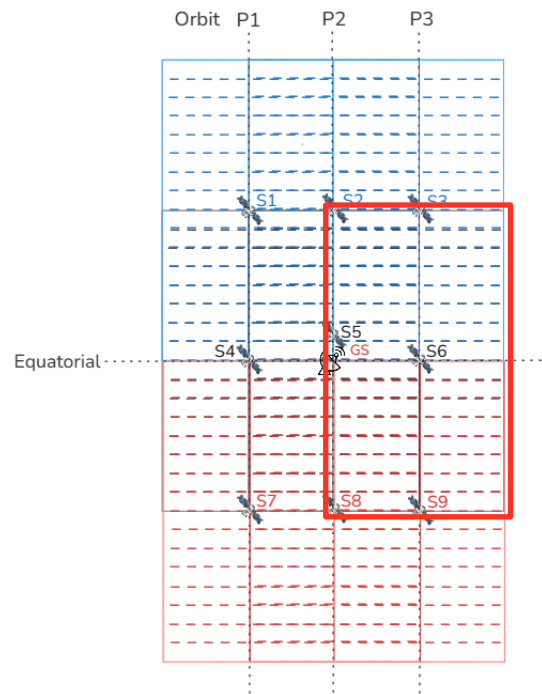
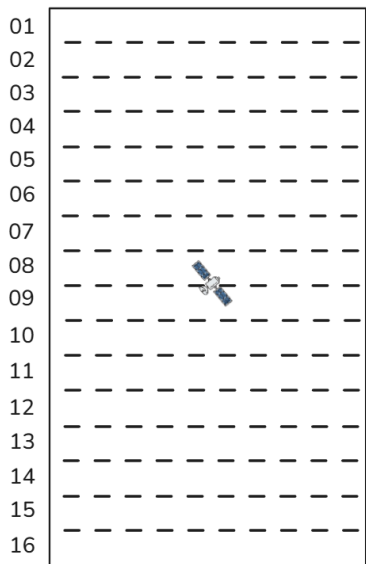
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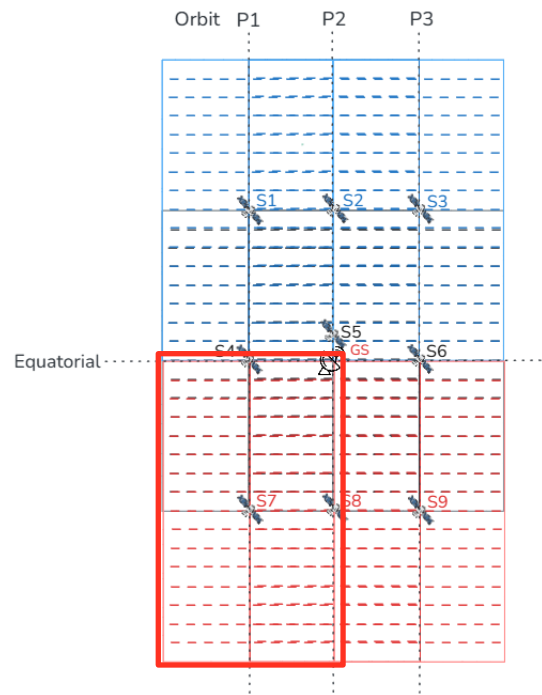
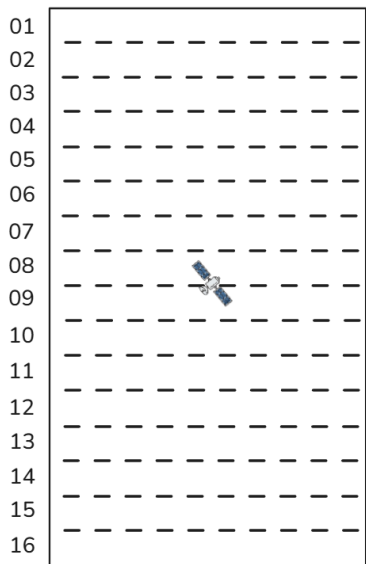
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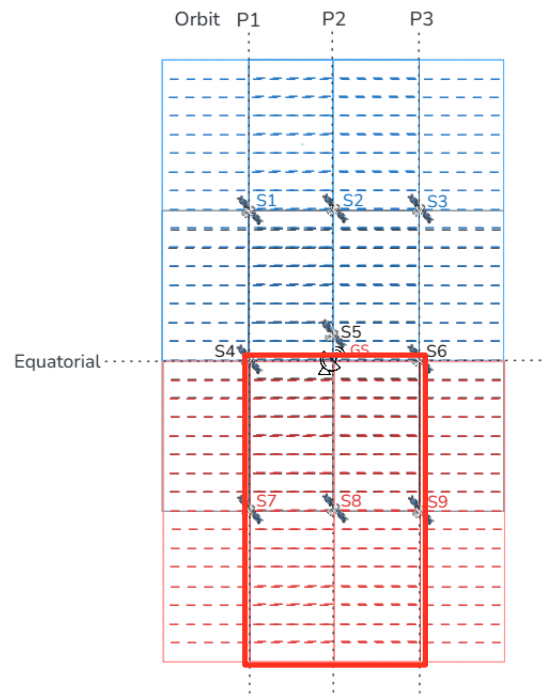
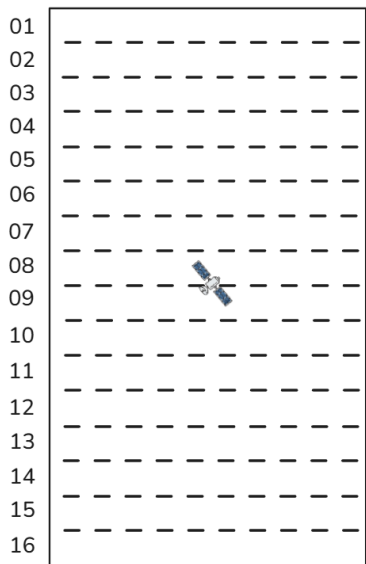
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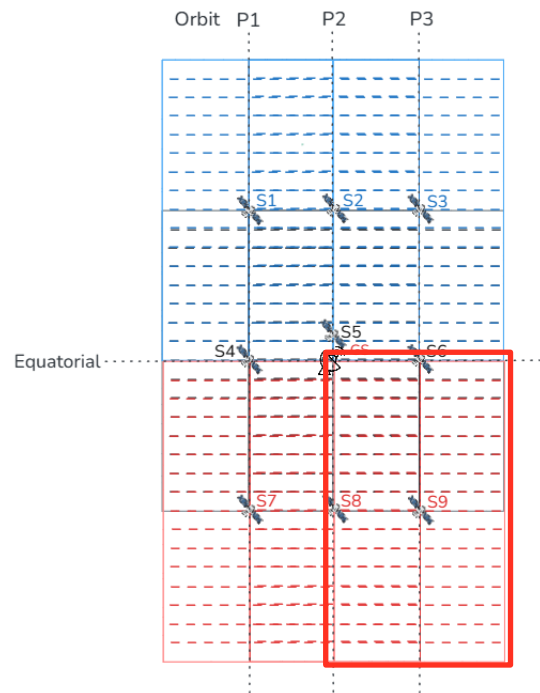
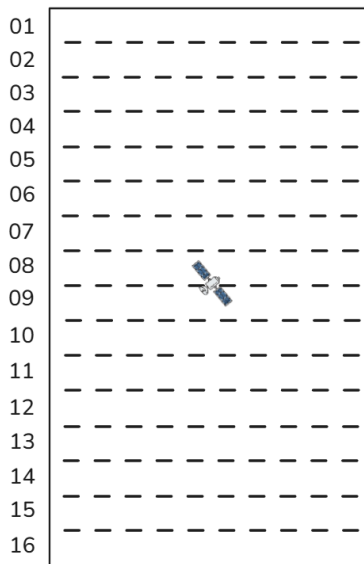
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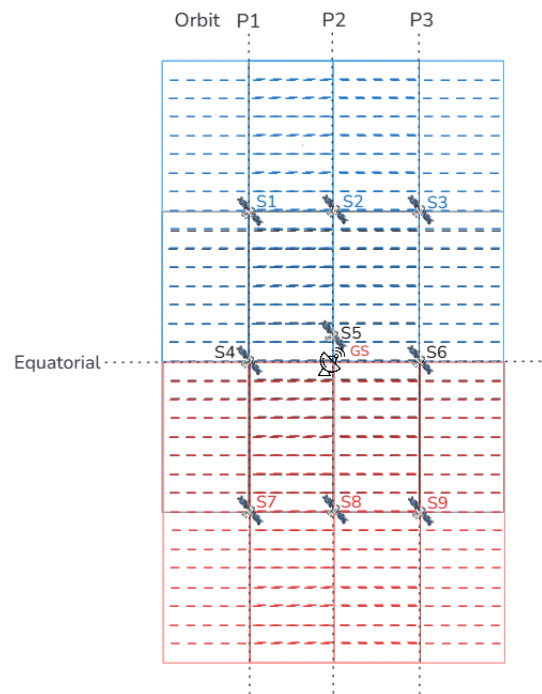
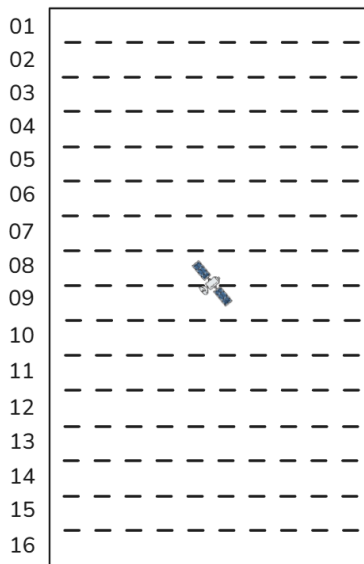
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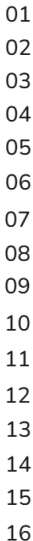


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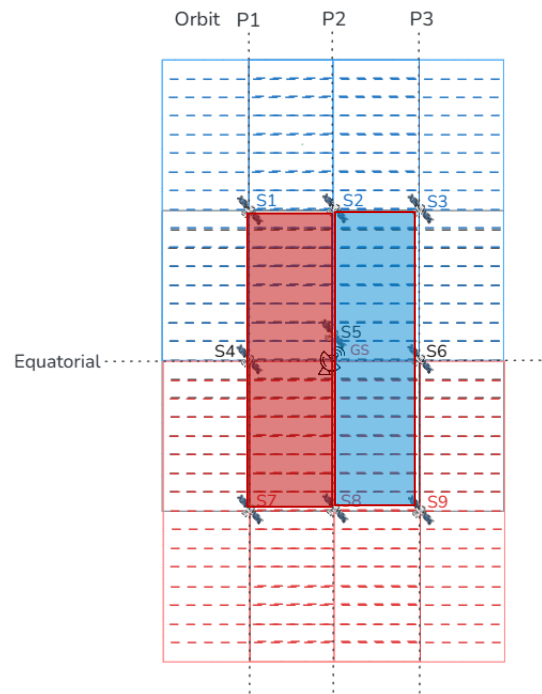
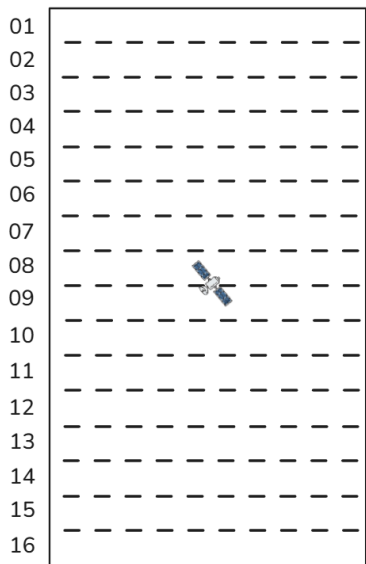


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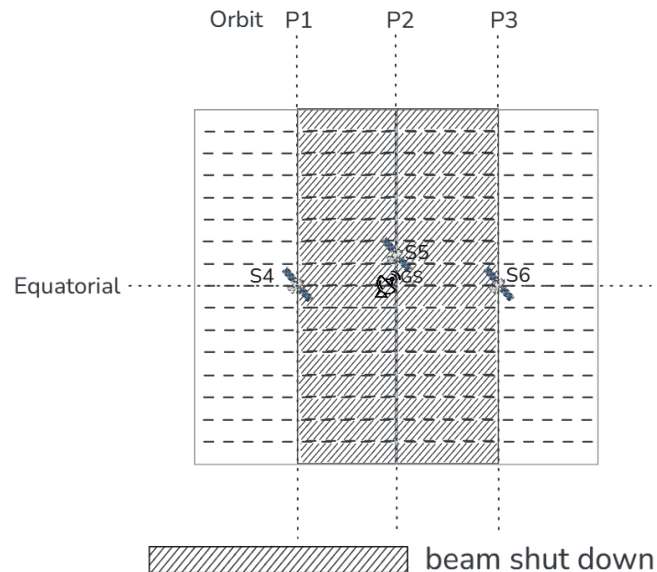
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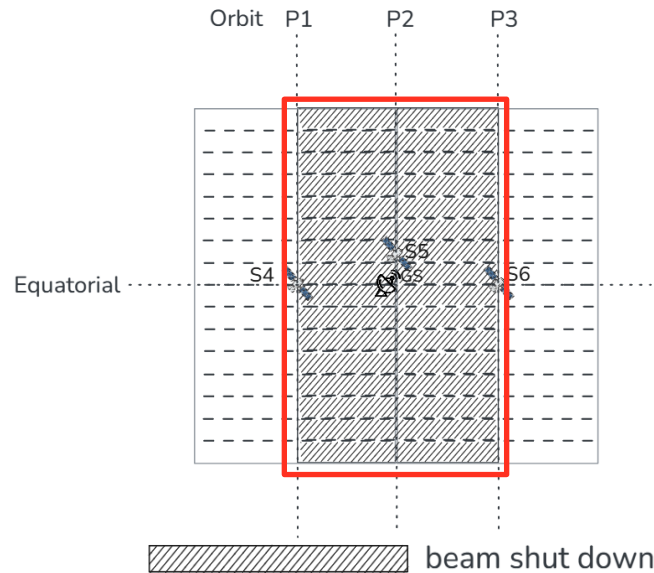
Method — Time-Slot Beam Role Record

- Critical satellite (S5) :
 - Aggregate EPFD exceeds the limit
 - Higher beam-level EPFD contributions are shut down first.
- Helper satellite (S4 、 S6) :
 - a neighboring visible LEO satellite whose beam footprints overlap with critical beams.



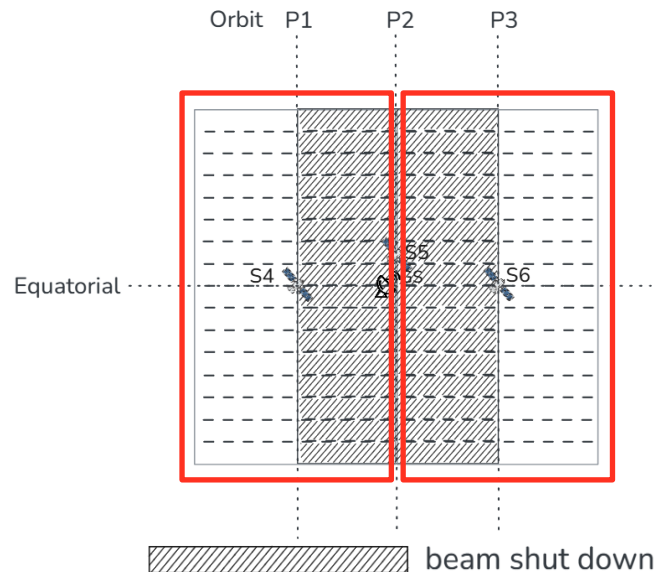
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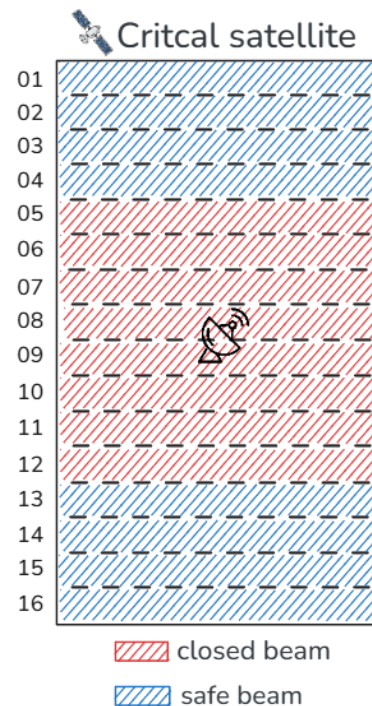
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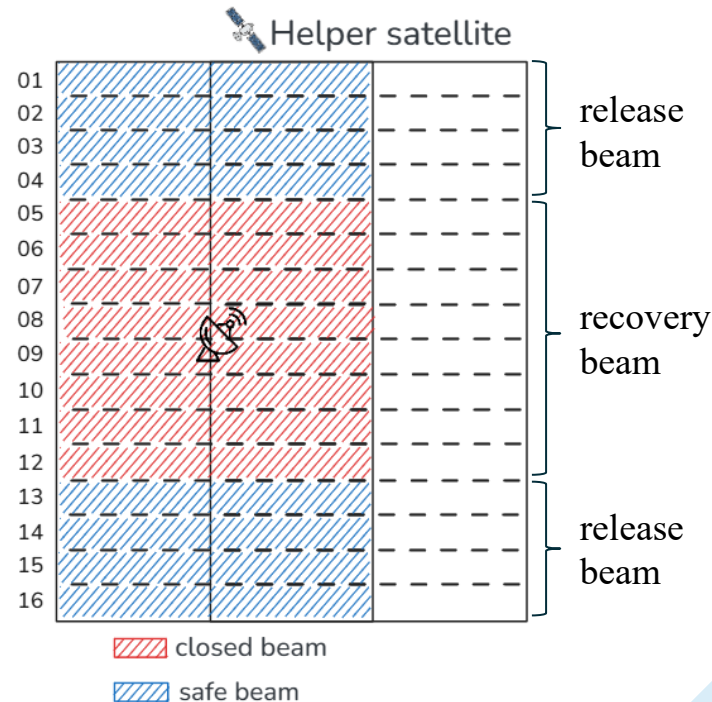
Method — Time-Slot Beam Role Record

- Critical-side beam roles:
 - closed beam
 - safe beam
- Helper-side beam roles:
 - recovery beam
 - release beam
- Footprint overlap relation:
 - Critical closed beams $\leftrightarrow$ Helper recovery beam
 - Critical safe beams $\leftrightarrow$ Helper release beam



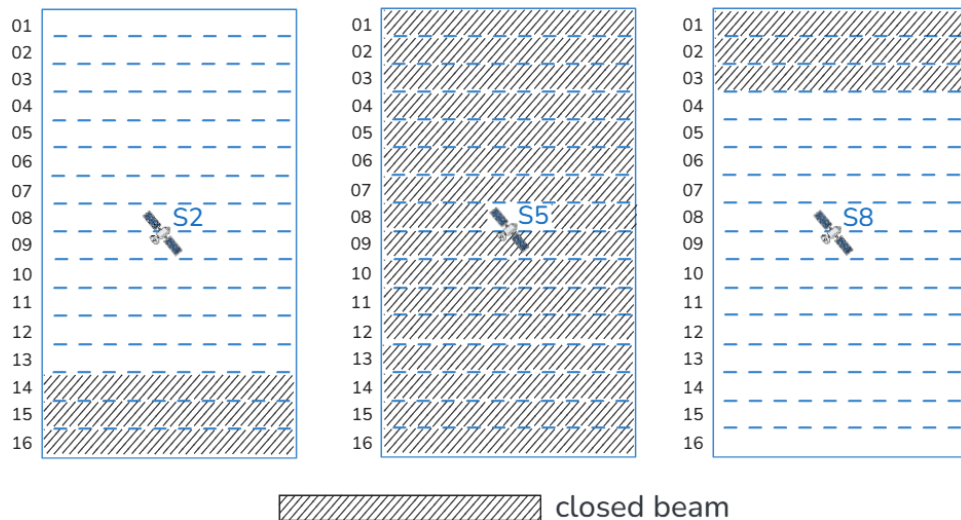
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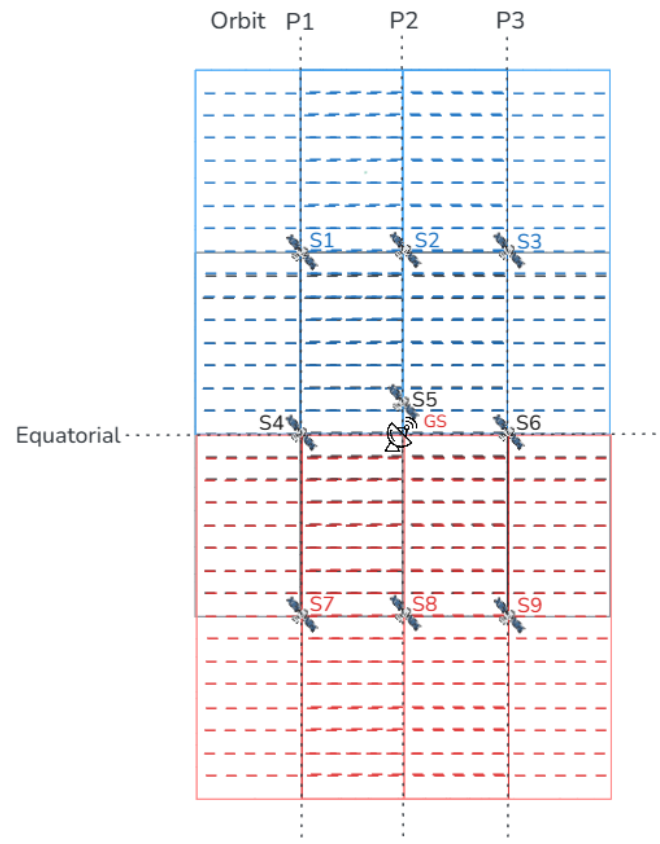


Method — Time-Slot Beam Role Record

- Critical satellite: S2 、 S5 、 S8



- Helper satellite: S1 、 S3 、 S4 、 S6 、 S7 、 S9



Method — Safe-release candidate graph

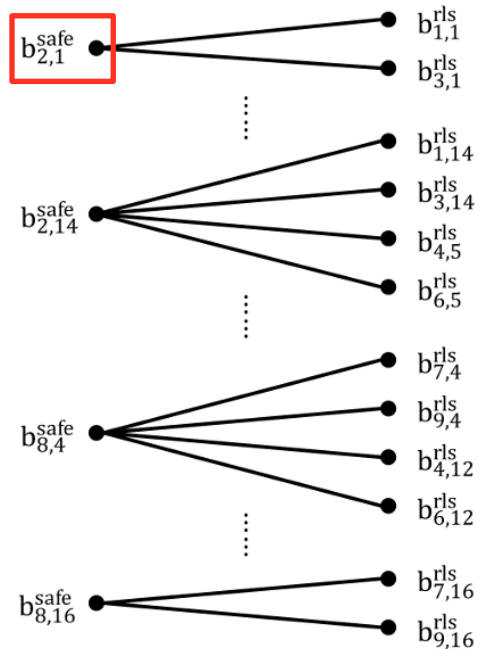
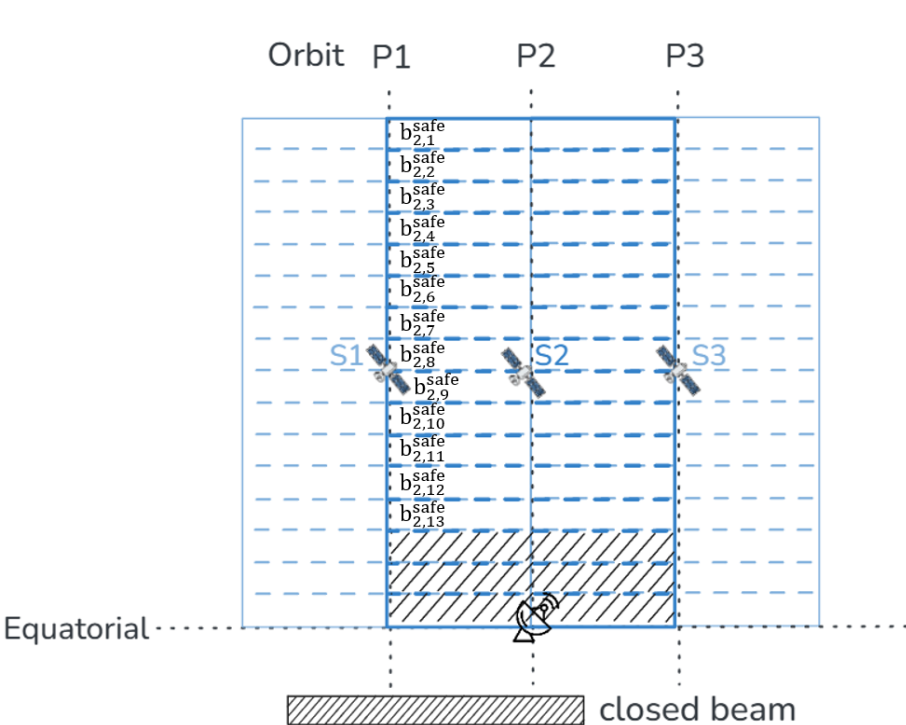


Fig. 3. Safe-release candidate graph between critical safe beams and helper release beam candidates. The left-side nodes $b_{s,\ell}^{safe}$ denote safe beams on critical satellites, and the right-side nodes $b_{h,k}^{rls}$ denote helper release beam candidates. An edge $e = (b_{s,\ell}^{safe}, b_{h,k}^{rls})$ exists when the two beams have footprint overlap.

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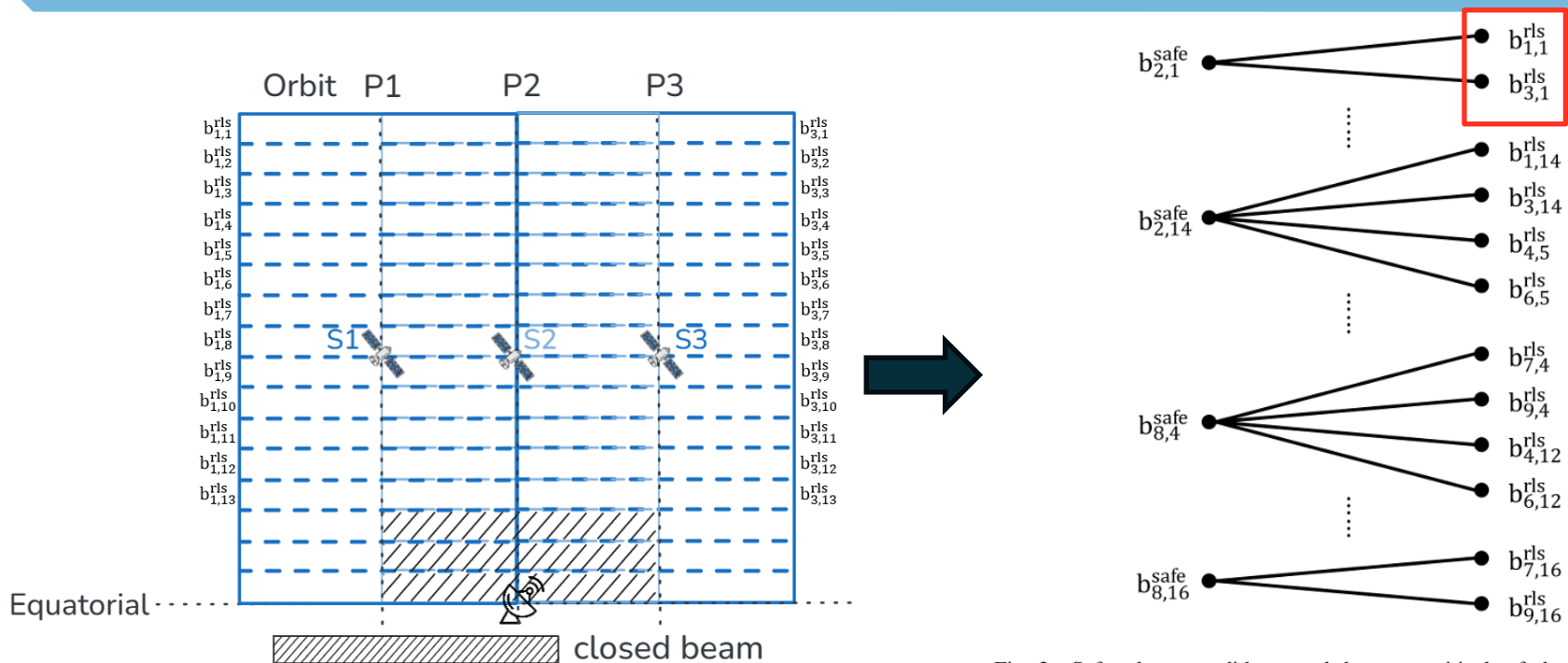


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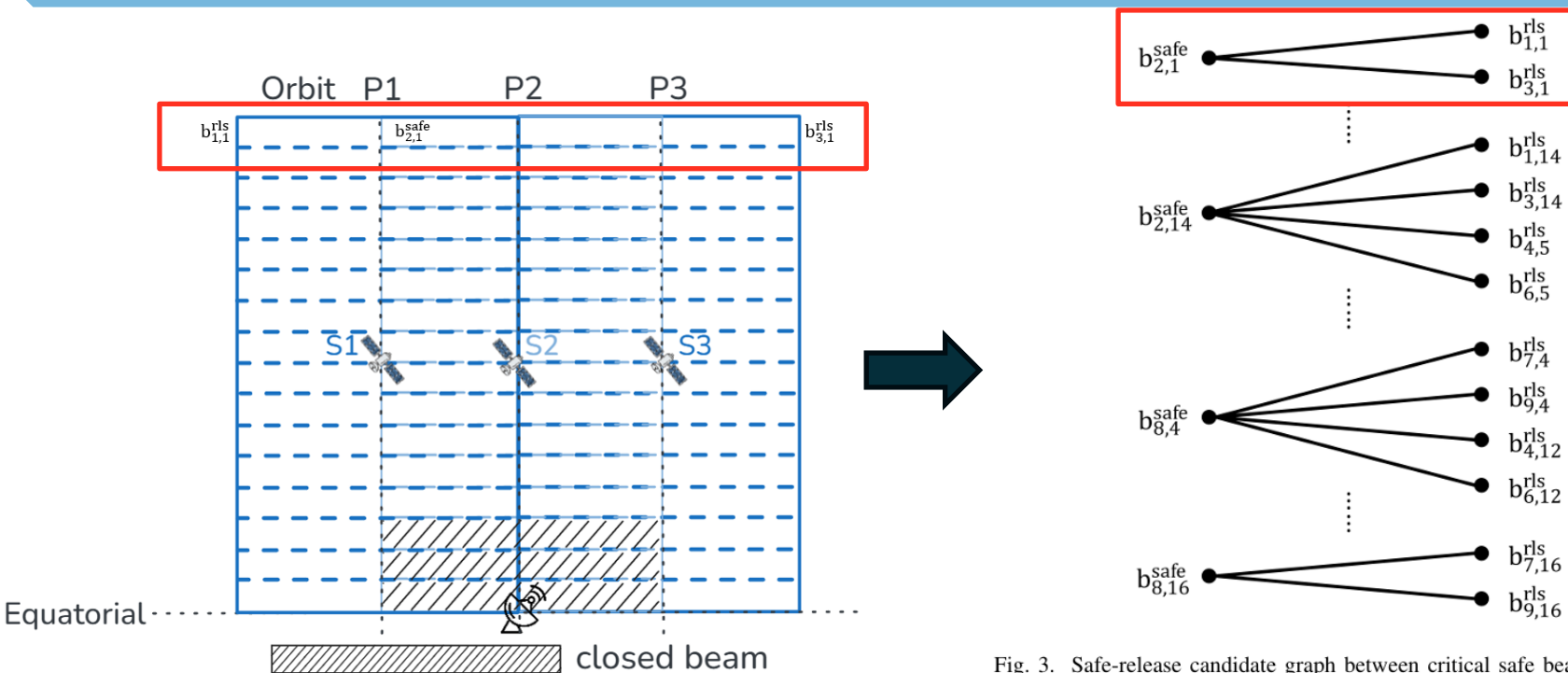


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Method — Closed- recovery candidate graph

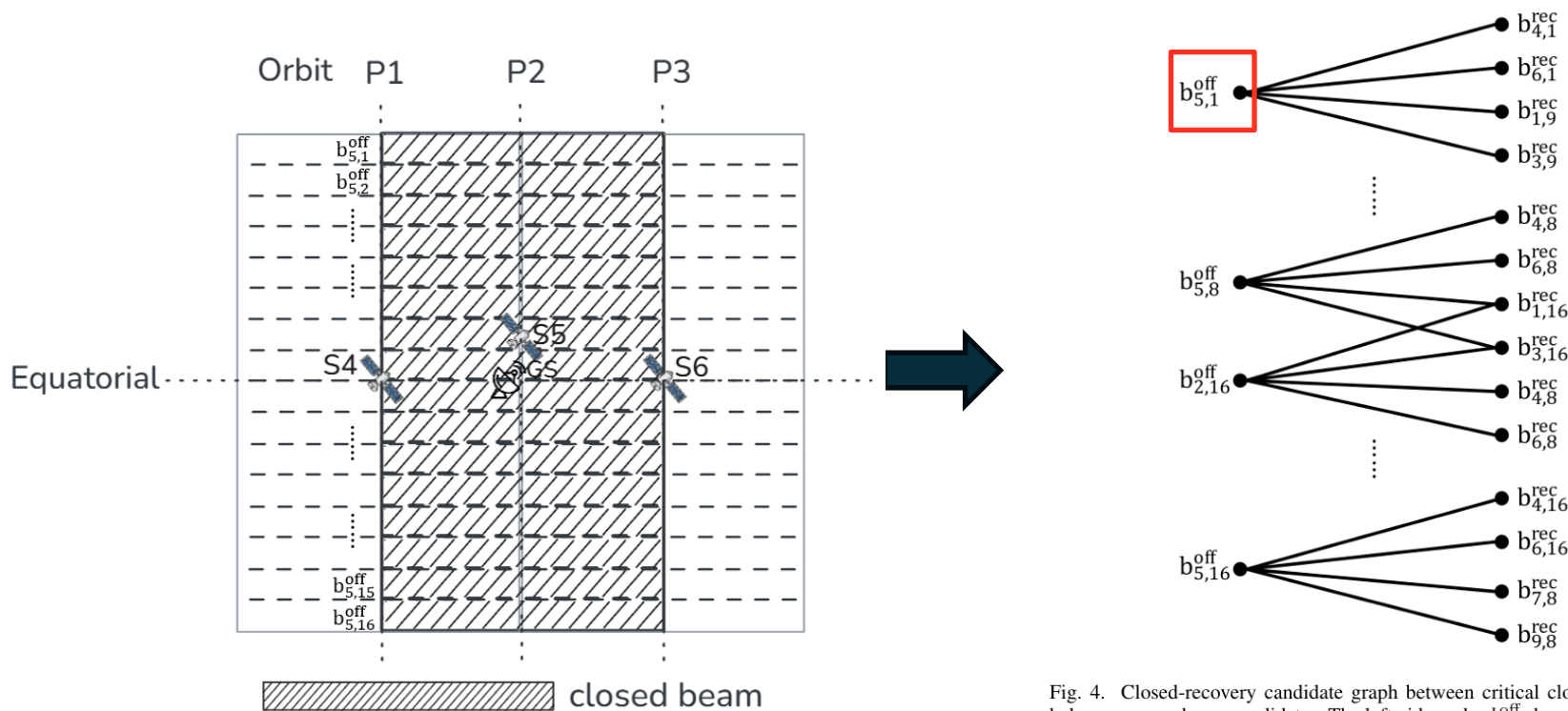


Fig. 4. Closed-recovery candidate graph between critical closed beams and helper recovery beam candidates. The left-side nodes $b_{s,\ell}^{off}$ denote closed beams on critical satellites, and the right-side nodes $b_{h,k}^{rec}$ denote helper recovery beam candidates. An edge $e = (b_{s,\ell}^{off}, b_{h,k}^{rec})$ exists when the two beams have footprint overlap.

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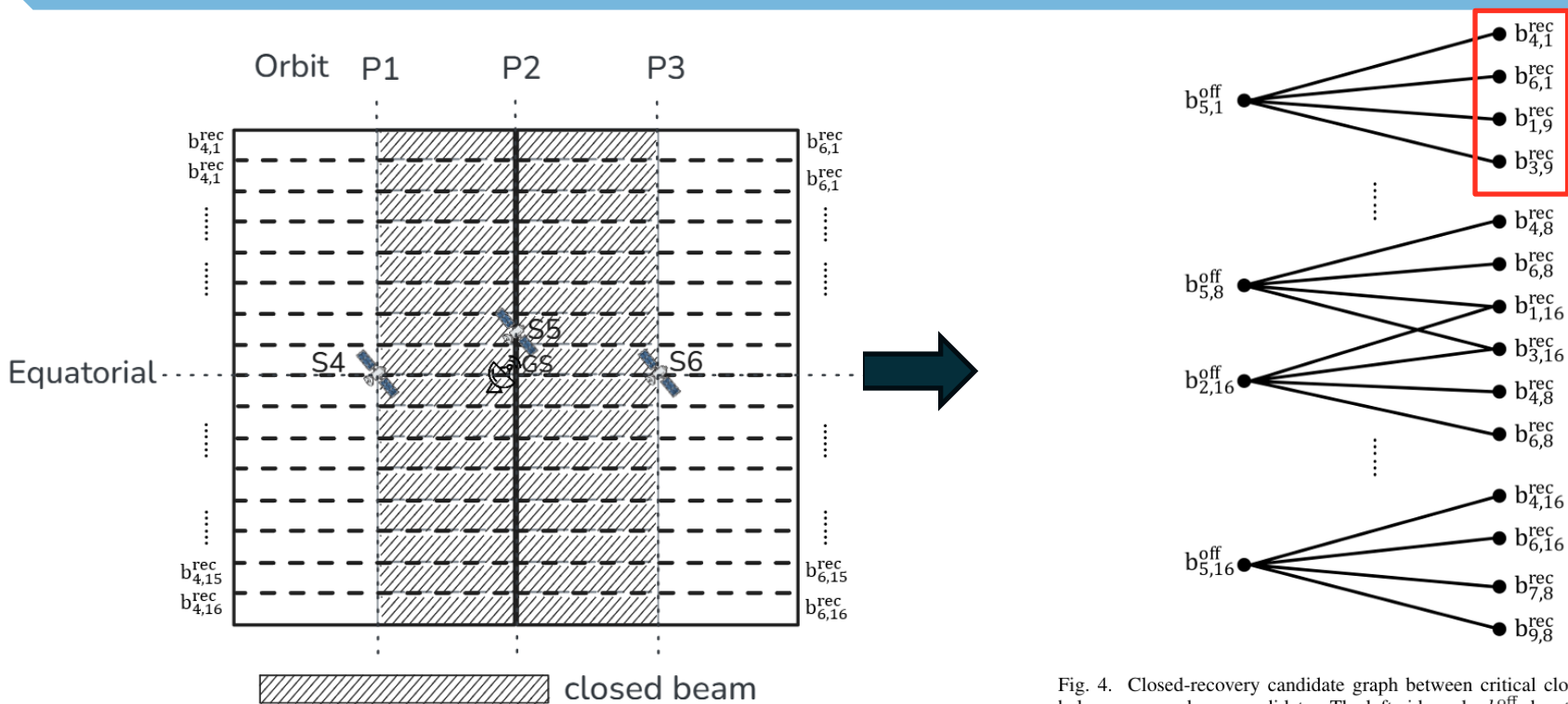


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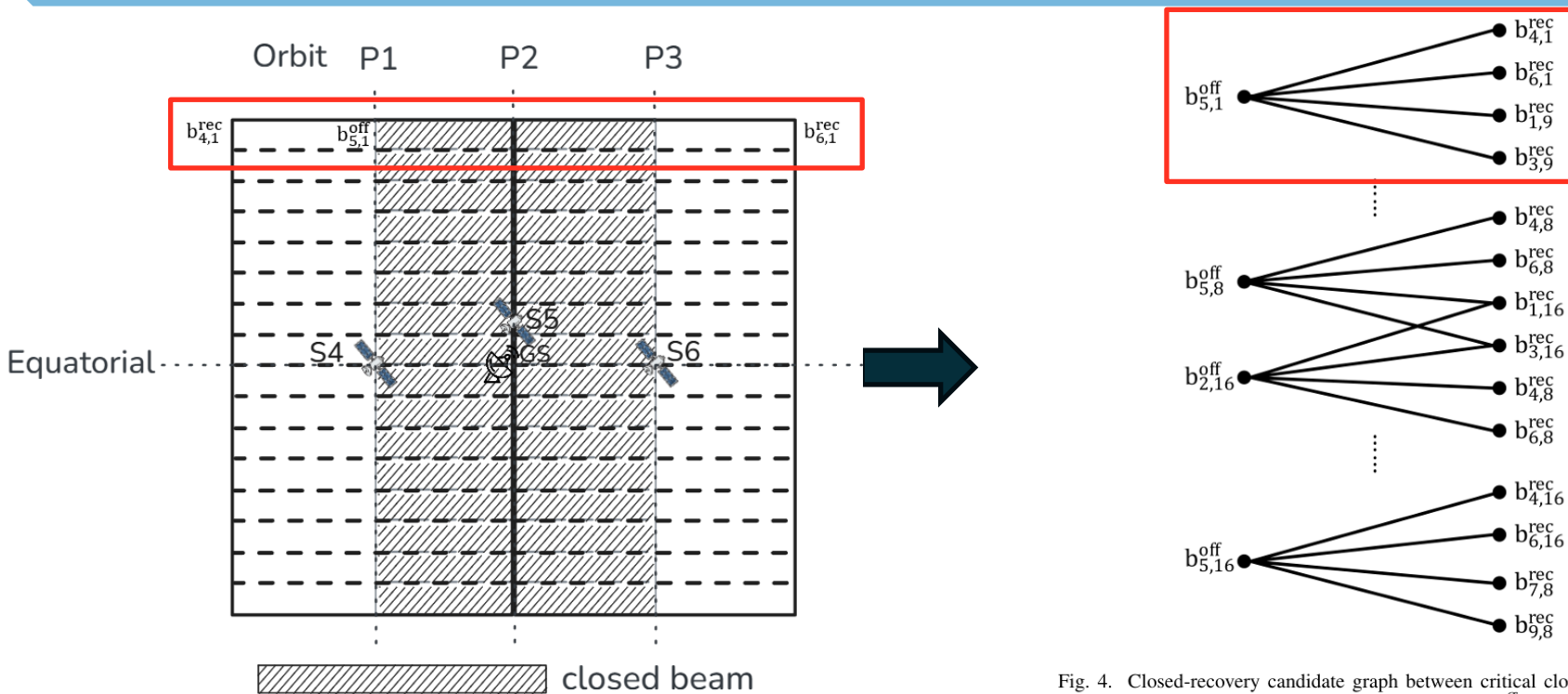


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Method — Safe-Beam Reassociation for Helper Users (SBR)

- Goal : Use critical safe beams to serve the users originally on helper release beams, so that the helper satellite can release available power

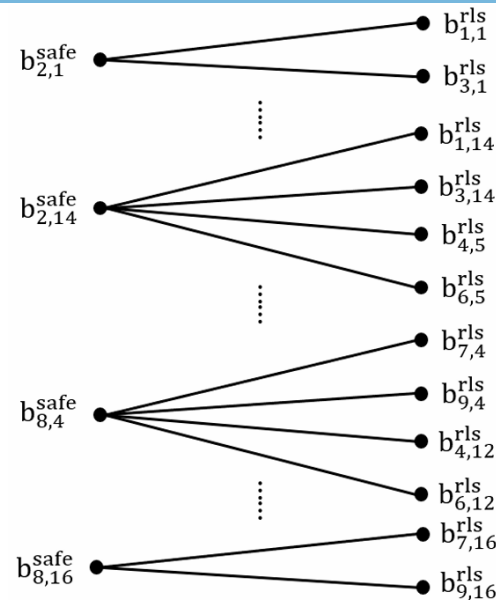
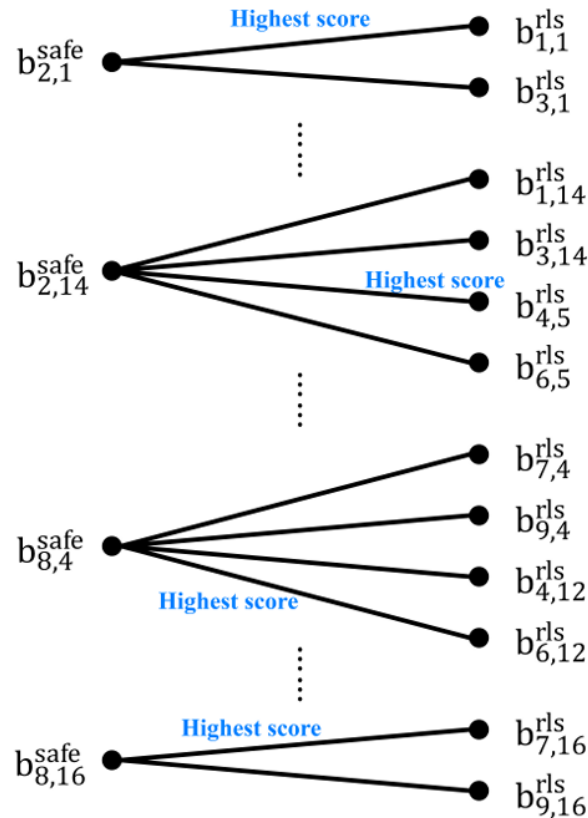


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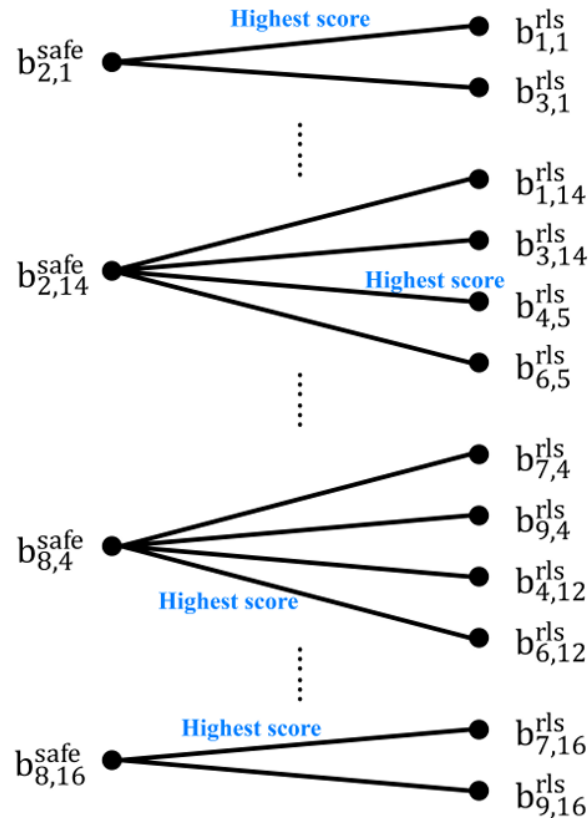
Method — Safe-Beam Reassociation for Helper Users (SBR)

- Release Score
 - Release score = maximum helper user load that can be reassociated under safe beam residual capacity.
 - Higher score means larger power-release benefit.
- The score reflects how much helper user can be reassociated to a critical safe beam.



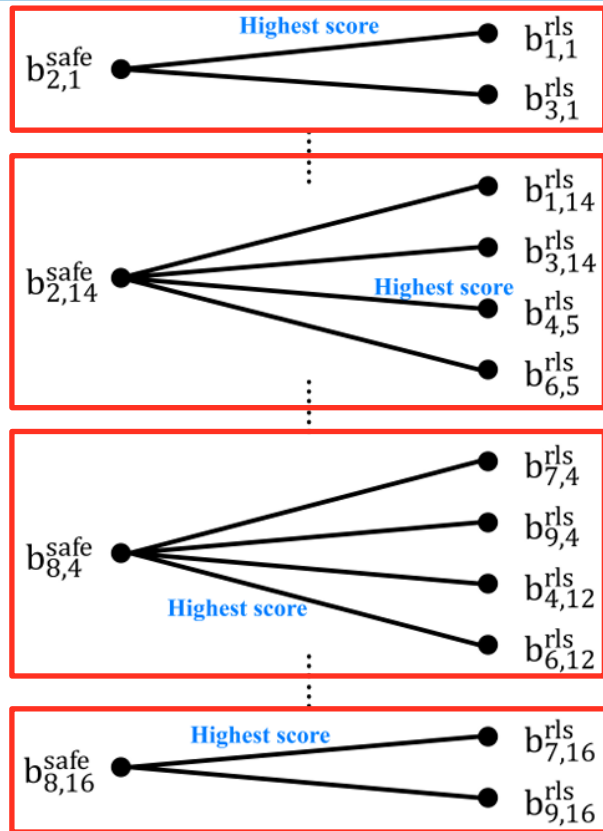
Method — Safe-Beam Reassociation for Helper Users (SBR)

- Divide the graph into connected components.
- Select the highest-score edge in each component.

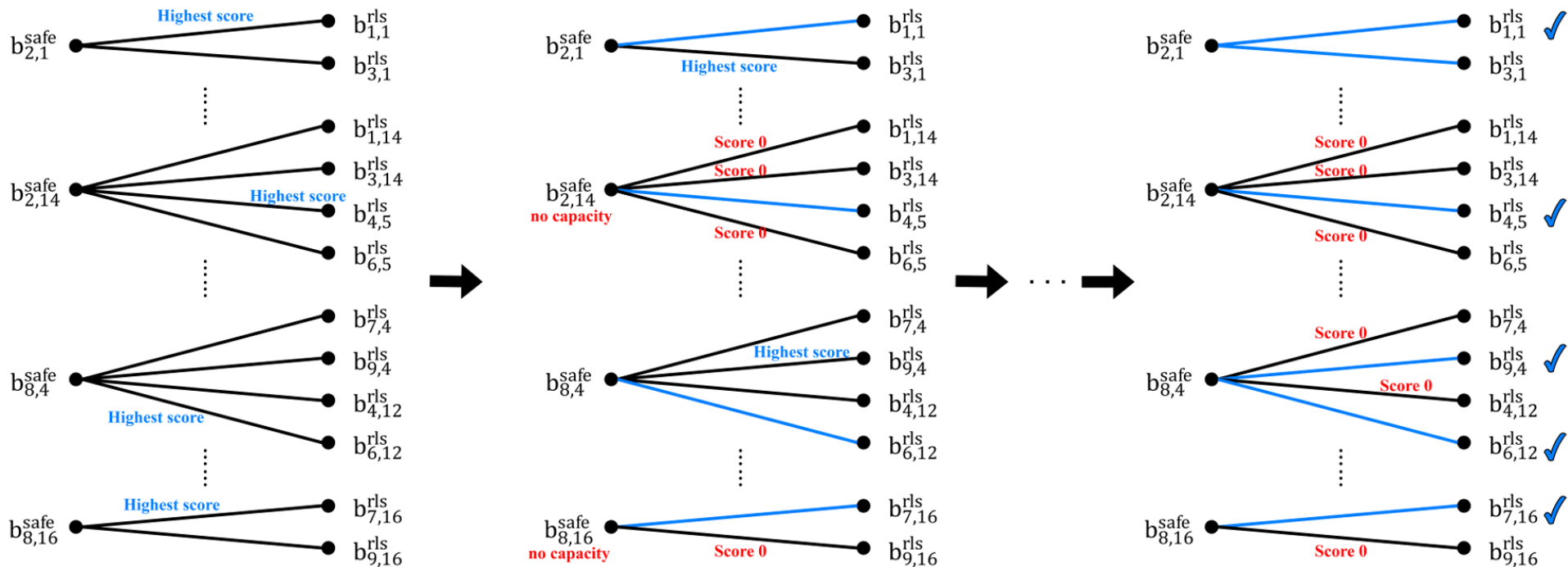


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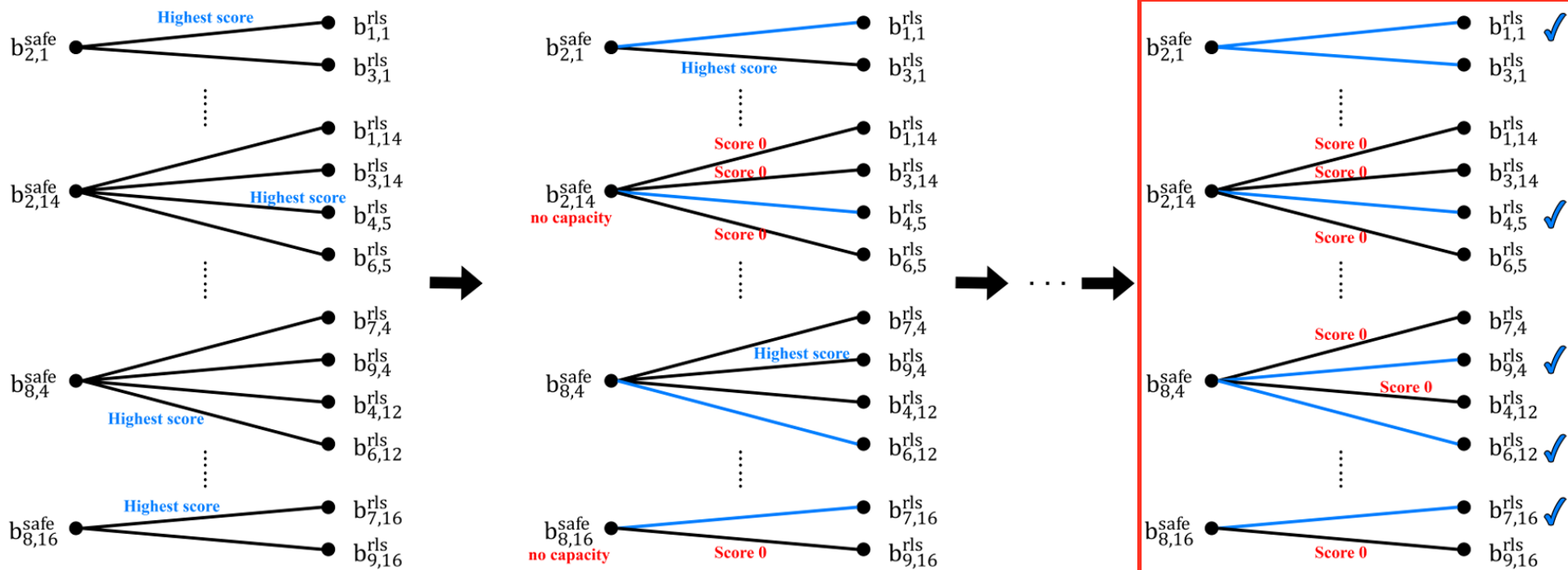


Method — Safe-Beam Reassociation for Helper Users (SBR)





Method — Safe-Beam Reassociation for Helper Users (SBR)



Score

Select Edge

Update

Method — Released Power Allocation

- SBR releases helper satellite transmit power.
- Released power is added back to the helper satellite's available power budget.
- Helper recovery beams are ranked by priority-weighted recovery demand.
- Released power is allocated in this order, boosting each helper recovery beam up to its transmit power limit.

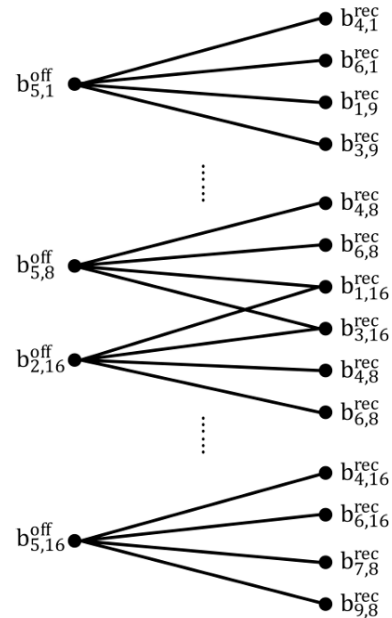


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Method — Safe-Beam Reassociation for Helper Users (SBR)

- Goal: Use helper recovery beams to serve users affected by critical beam shutdown.

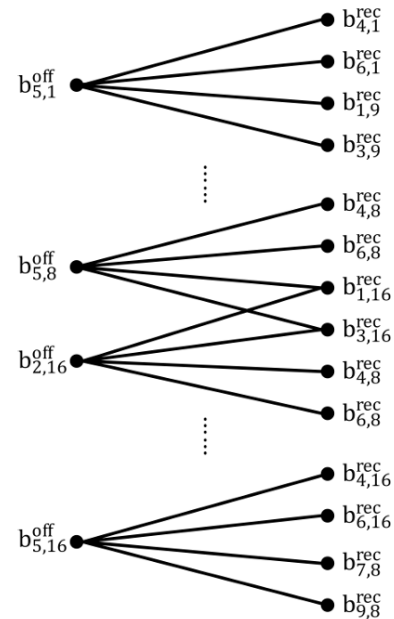
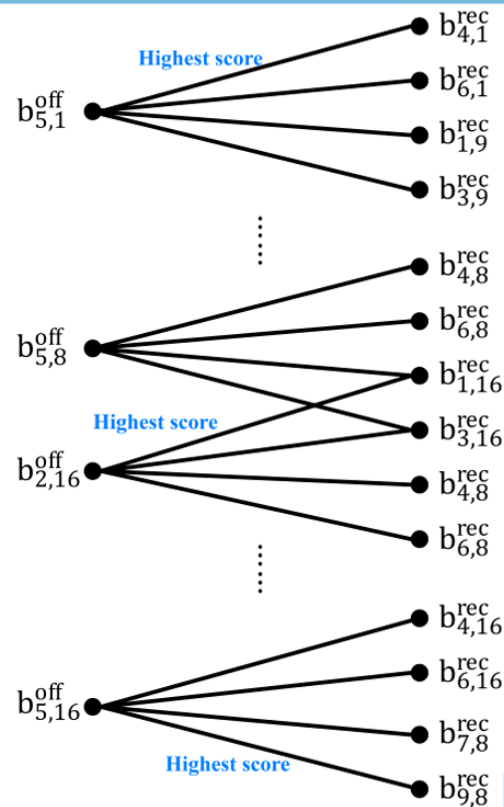


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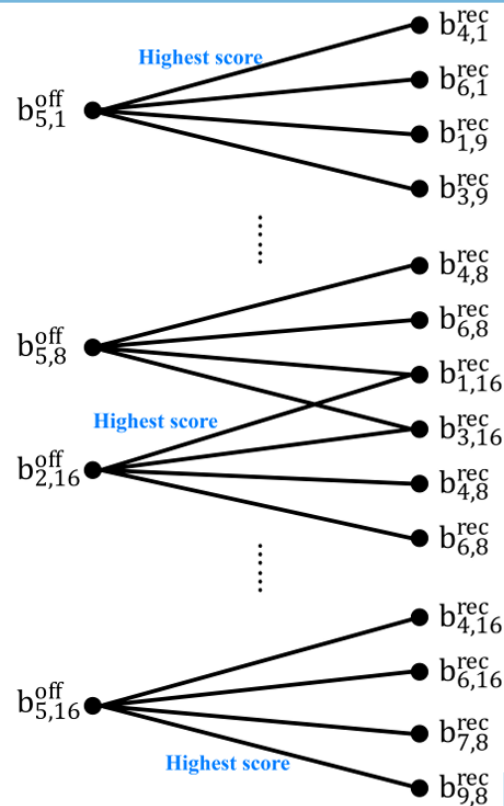
Method — Helper-Beam Reassociation for Closed-Beam Users (HBR)

- Recovery Score:
 - For each closed-recovery edge, sort candidate users by:
 - Service priority: high to low
 - Required capacity: low to high within the same priority
 - Estimate the recoverable user subset under the remaining capacity of the helper recovery beam.
 - Recovery score = sum of each estimated user's priority weight $\times$ its satisfaction.



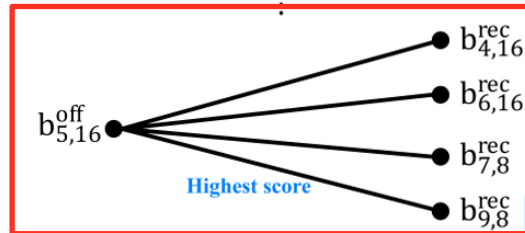
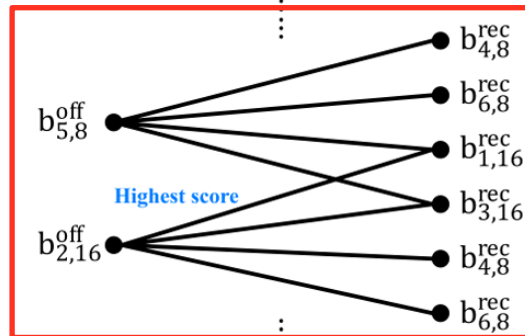
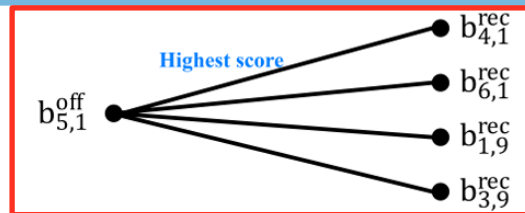
Method — Helper-Beam Reassociation for Closed-Beam Users (HBR)

- Divide the graph into connected components.
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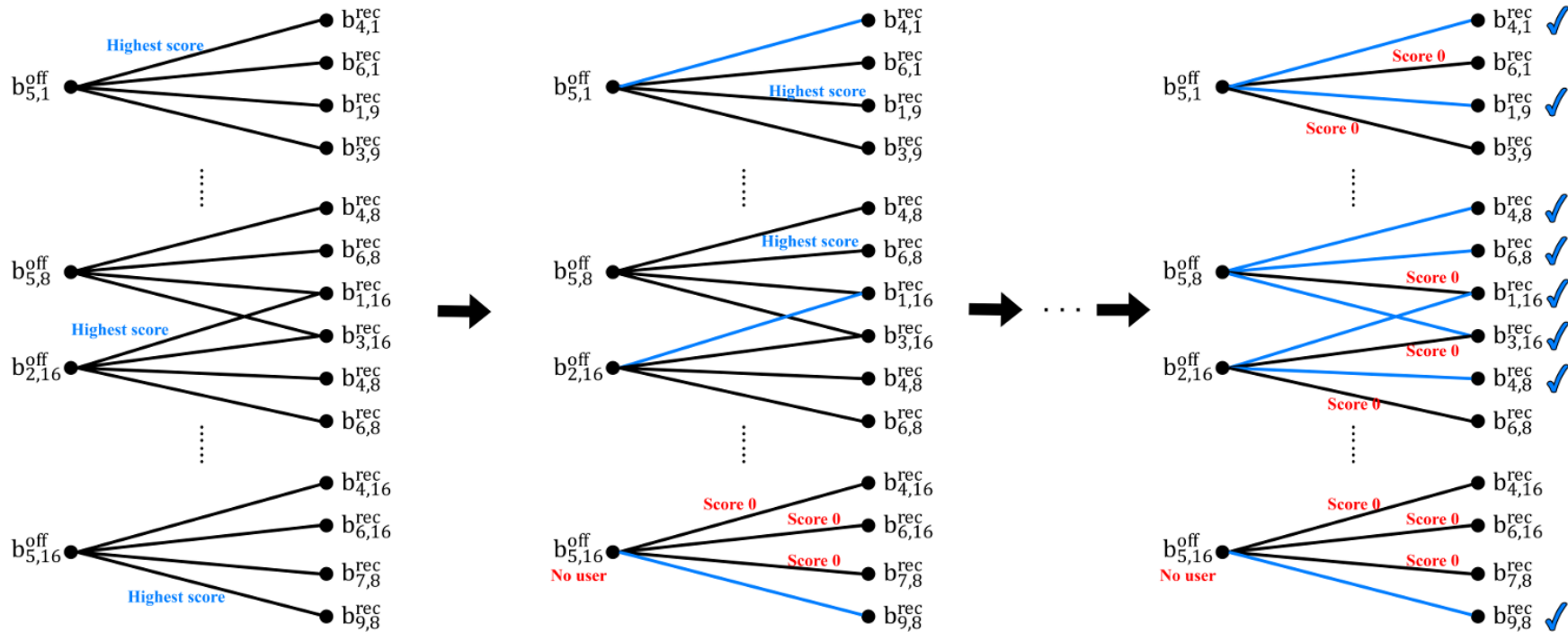
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- Select the highest-score edge in each component.





Method — Helper-Beam Reassociation for Closed-Beam Users (HBR)



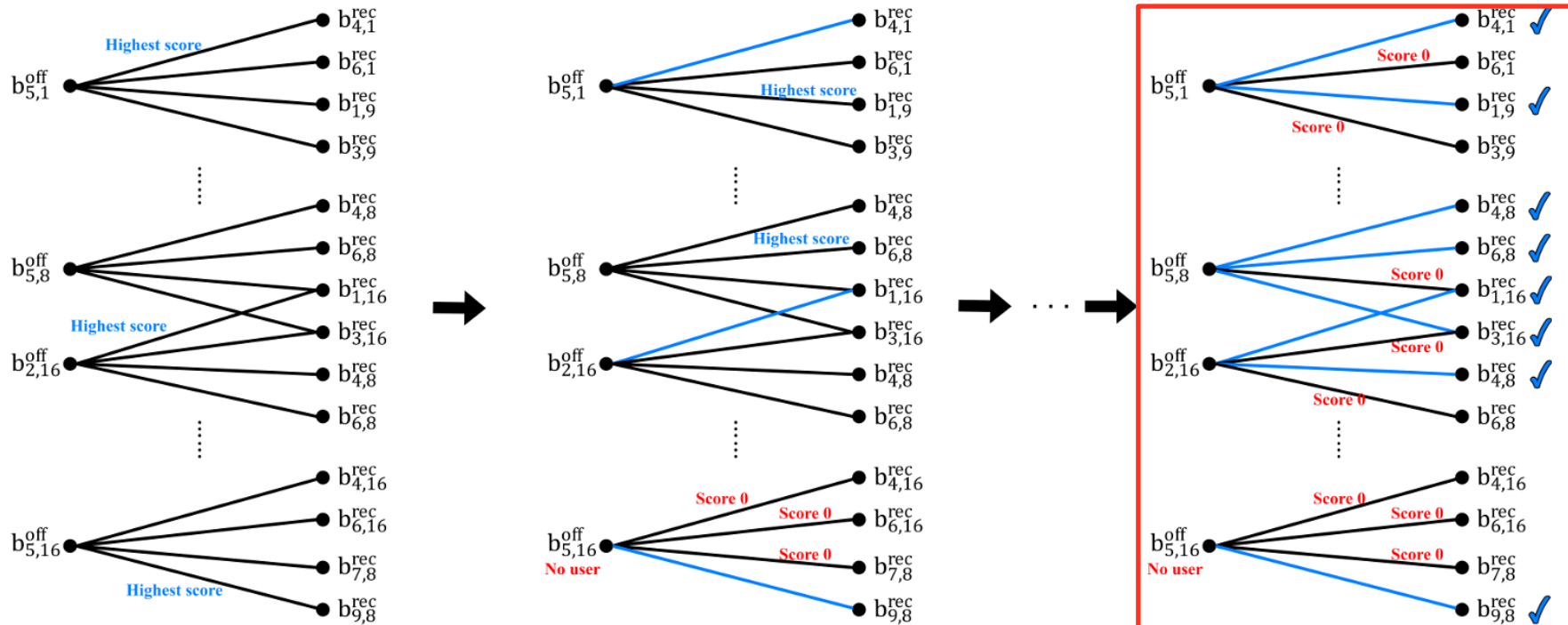
Score

Select Edge

Update



Method — Helper-Beam Reassociation for Closed-Beam Users (HBR)



Score

Select Edge

Update

Evaluation — Simulation Setup

- Tool Used : STK (Satellite Tool Kit) and Matlab

Parameter	Value	Description
LEO Constellation	OneWeb-like	Simulation setting
Orbit Altitude	1200 km	Circular LEO orbit
Orbit Plane Inclination	87.9°	Near-polar Walker Star
Beams per Satellite	16	Fixed Ku-band footprint layout
Simulation Time Window	2025/12/16 12:11:00–12:13:30	Critical observation period during the critical satellite pass
Time Slot	1 s	Time-step interval
Number of GSO and GS	1	Co-frequency GSO and GSO ground station
GSO Location	(0°N, 30.6°E)	GSO Latitude and longitude
GS Location	(0°N, 30.6°E)	Same Location as the target GSO satellite
Number of users	30, 50, 70	Different user load under each LEO satellite
UE Mobility	Static	UE positions remain static

Evaluation — Four Strategy

Strategy	Method Description
Beam shutdown only	Shuts down high-interference beams on the critical satellite to satisfy the EPFD limit
PC + Tilt	Reduces interference by adjusting transmit power and satellite tilt angle
Only HBR	Uses helper recovery beams to serve users under closed beams with default helper-satellite power
EABR	The proposed method

Evaluation — Agg EPFD and number of closed beams

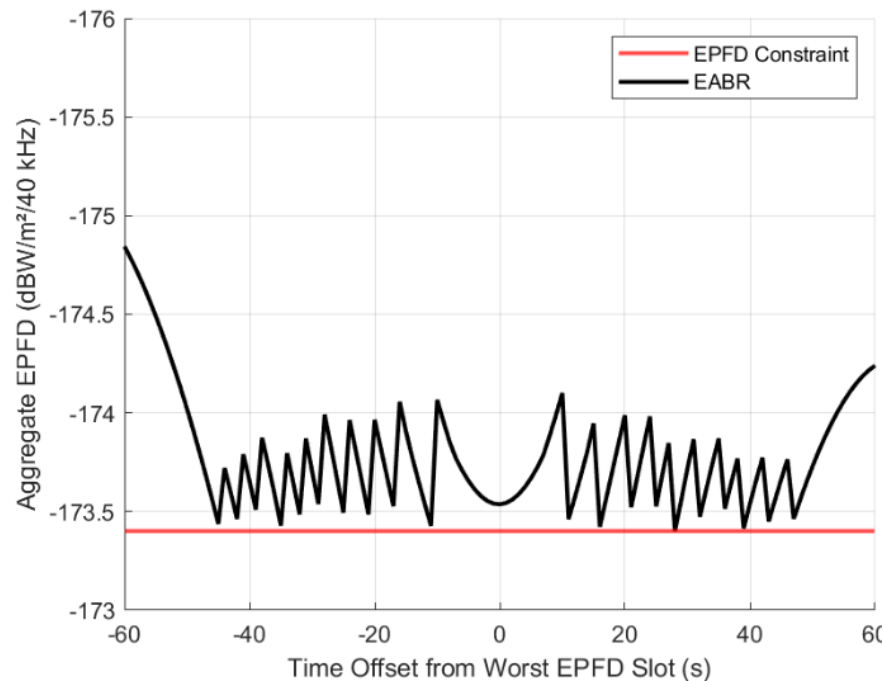


Fig. 4. Aggregate EPFD over time during the critical period.

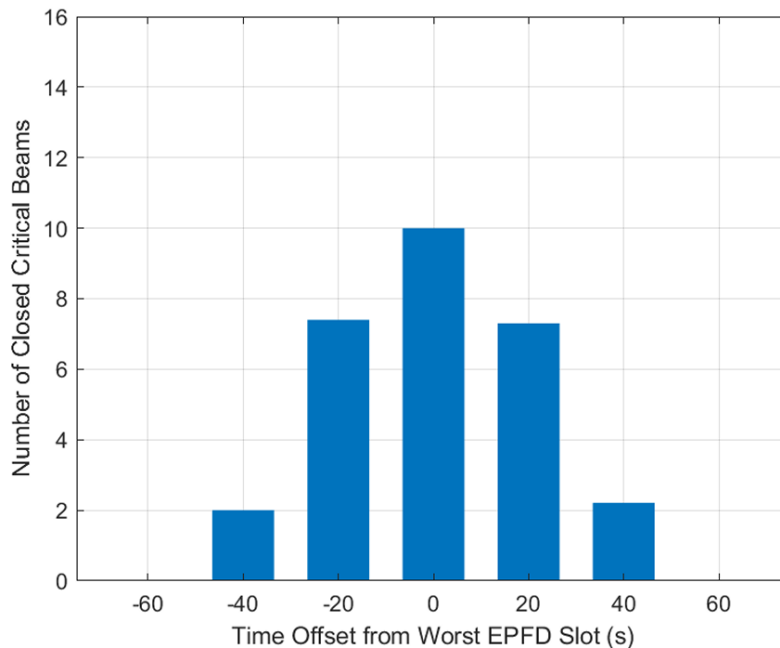
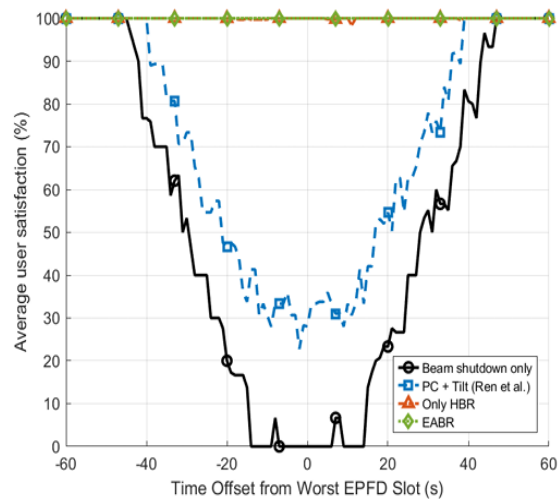
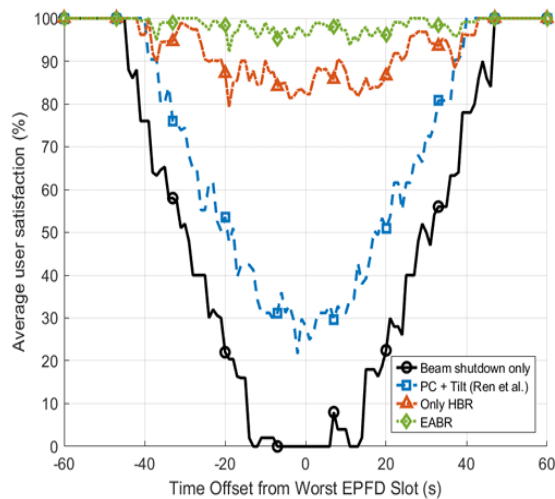


Fig. 5. Number of closed beams on critical satellites over time during the critical period.

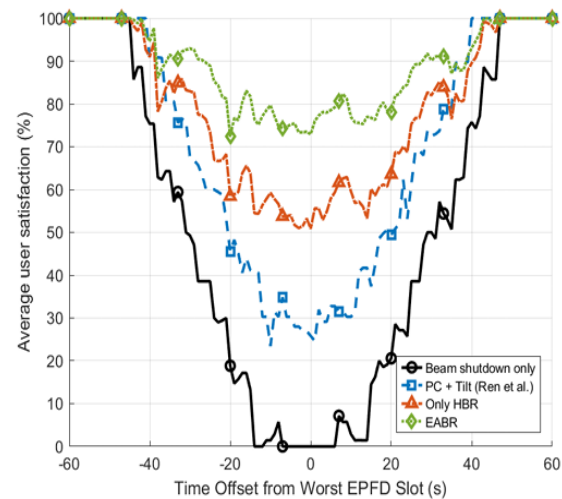
Evaluation — Average user satisfaction



(a)



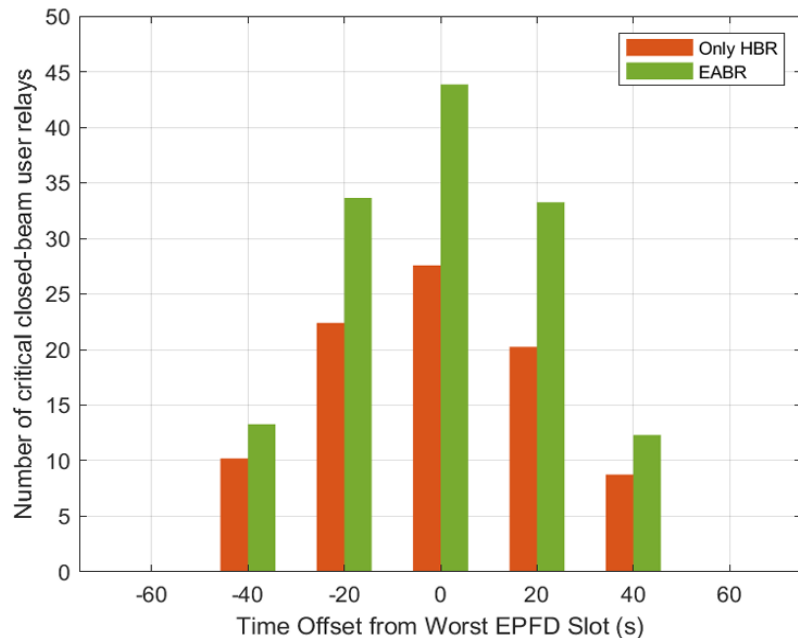
(b)



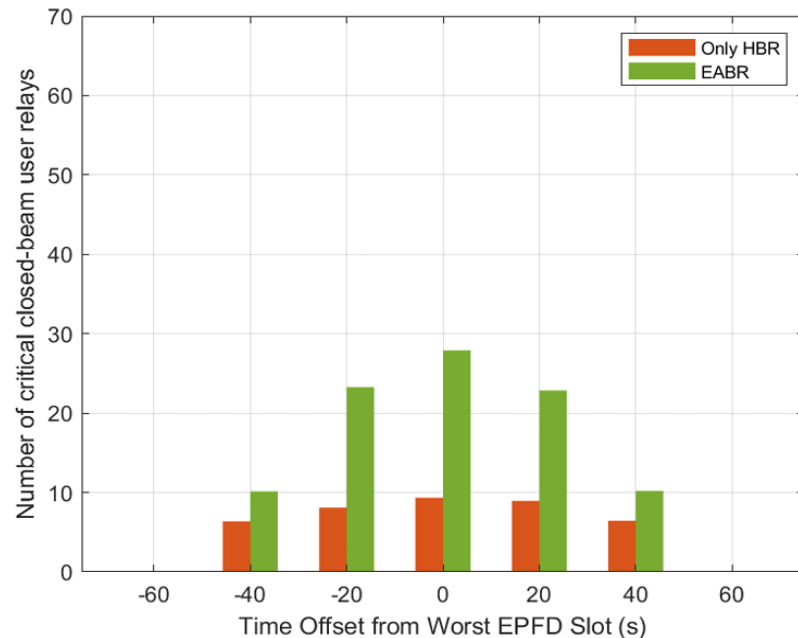
(c)

Fig. 6. Average user satisfaction of different interference mitigation methods under different user loads. (a) 30 users. (b) 50 users. (c) 70 users.

Evaluation — Number of relayed critical users



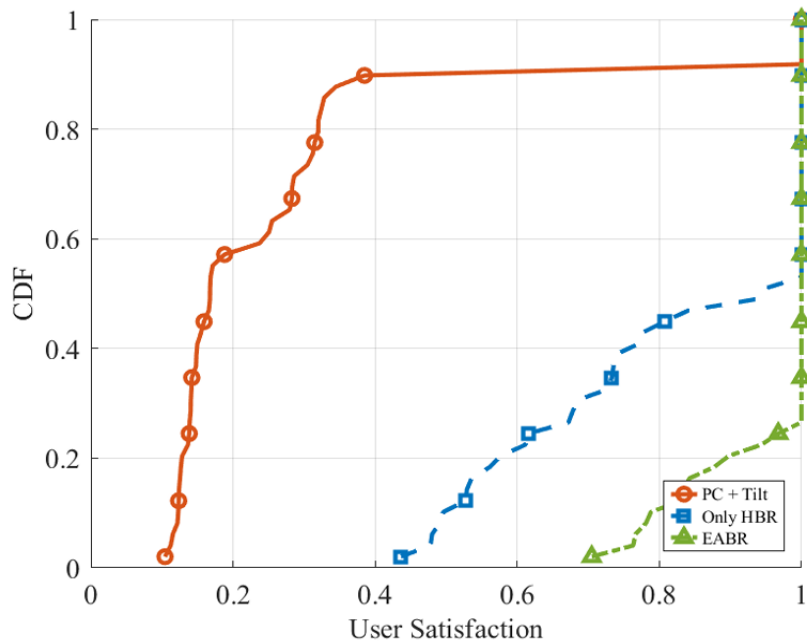
(a)



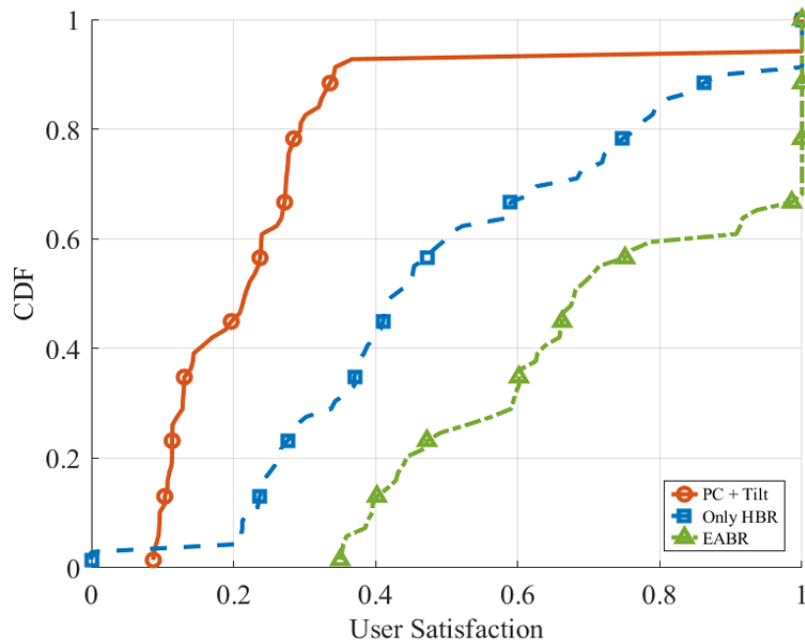
(b)

Fig. 7. Number of recovered closed-beam users with and without SBR under different user loads. (a) 50 users. (b) 70 users.

Evaluation — Satisfaction CDF



(a)



(b)

Fig. 8. CDF of user satisfaction for different interference mitigation methods under different user loads. (a) 50 users. (b) 70 users.

Evaluation — Different EPFD limit

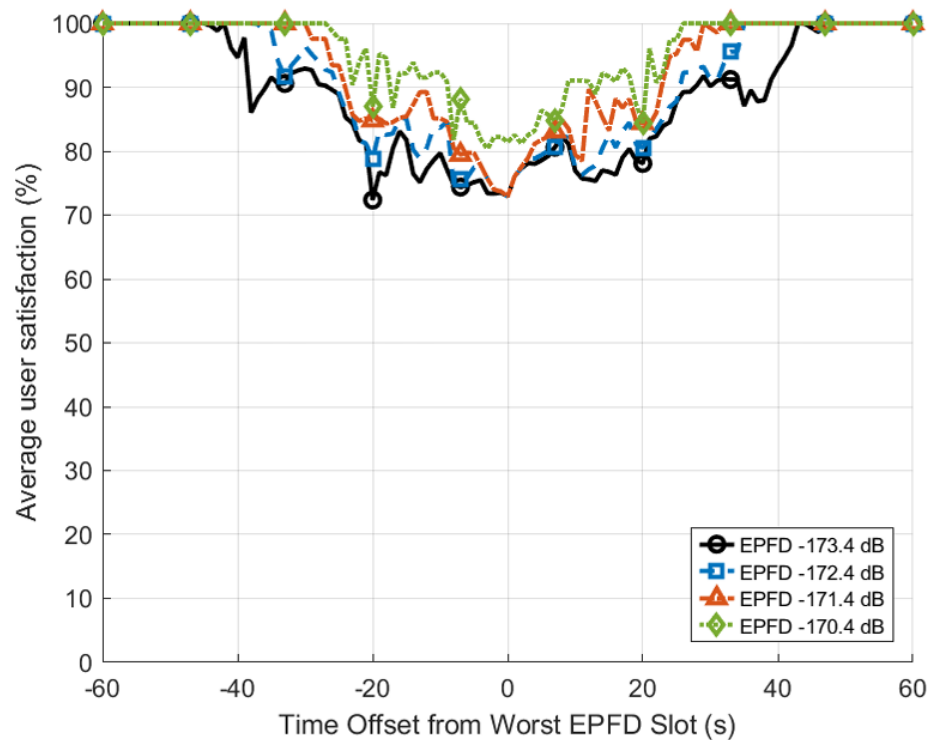


Fig. 9. Average user satisfaction of EABR under different EPFD limits.

Conclusion

- This thesis proposed EPFD-Aware Beam Reassociation(EABR) for Service Recovery, a service recovery framework for multi-beam LEO satellite systems under NGSO–GSO coexistence.
- The proposed framework uses time-slot beam role records, safe-beam reassociation, released power allocation(SBA) , and helper-beam reassociation(HBA) to recover users affected by beam shutdown.
- Simulation results show that the proposed framework keeps the aggregate EPFD below the required limit and achieves higher user satisfaction than baseline methods.



Thank you for listening.